

Solidity Code Explanation



```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract MyFavoriteNumber {
    uint public myNumber = 10;
    // This is our new function!
    function setMyNumber(uint _newNumber) public {
        myNumber = _newNumber;
    }
} Explain this code in details and also make notes related to this
```

Alright! Let's break down this Solidity code **line by line**, then I'll give you **clear notes** you can keep for quick revision.

✓ The Code

```
solidity

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    }
}
```

🔍 Detailed Explanation

📄 // SPDX-License-Identifier: MIT

- **What it does:**
This is a **software license identifier**.
- **Why it's used:**
Tells people how they can use your code. Here, **MIT License** means anyone can reuse, copy, modify it freely, but must include your license.
- **Good habit:**
Required by Solidity compiler (since Solidity 0.6.8) for clarity.

1 2
3 4 `pragma solidity ^0.8.20;`

- **What it does:**
Tells the compiler which version of Solidity to use.
 - **^ means:**
Compatible with version **0.8.20 and above**, but **below 0.9.0**.
 - **Why:**
Prevents errors due to version mismatches.
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 **contract** MyFavoriteNumber { ... }

- **What it is:**
Defines a **smart contract** — similar to a **class** in other languages.
 - **Name:**
MyFavoriteNumber is the contract's name.
 - **Purpose:**
Stores and allows changing your favorite number on the blockchain.
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 **uint public myNumber = 10;**

- **uint:**
Stands for **unsigned integer** — only positive numbers (whole numbers).
 - **public:**
Makes myNumber accessible from outside. Solidity **automatically creates a getter function** for public state variables.
 - **= 10:**
Initial value is set to 10.
-

 **function setMyNumber(uint _newNumber) public { ... }**

- **What it is:**
A **public function** to change the value of myNumber.
 - **uint _newNumber:**
Takes a **new number** from the user.
 - **public:**
Anyone can call this function.
 - **Inside:**
`myNumber = _newNumber;`
Updates the state variable with the new value.
 - **Impact:**
Saves the new number **permanently** on the blockchain.
-

Notes for Revision

✓ Key Points

Concept	Explanation
SPDX License Identifier	Required for open-source clarity. Here: MIT License.
Pragma	Defines compiler version compatibility (^0.8.20).
Contract	Basic building block of Ethereum apps. Like a class.
State Variable	<code>uint public myNumber</code> → stored on blockchain permanently.
Visibility: public	Variable & function are accessible from outside.
Setter Function	<code>setMyNumber</code> lets anyone update the value.
Getter	Solidity auto-generates a getter for public variables.

💡 How This Works On Blockchain

- When you **deploy** this contract, `myNumber` will be **10** by default.
- When you call `setMyNumber(25)`, `myNumber` is changed to **25**.
- This update is **permanent** and stored on Ethereum's blockchain.
- Anyone can read `myNumber` without cost. Changing it **costs gas**.

📁 When to Use This

- Learning how to **store and change state**.
- Understanding **state variables, visibility, and functions**.
- Basic building block for more complex smart contracts.

If you'd like, I can also make this into a **PDF note**, a **Markdown file**, or a **one-page cheat sheet**. Just tell me! 📄 ✨